Deep Reinforcement Learning for Analog Quantum Control: Robustness, Feedback, and Runtime Limits

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Reinforcement learning has been proposed as a unified route to analog quantum gate design, in which accuracy, leakage suppression, runtime and robustness against stochastic control errors are optimized jointly. We report an independent reimplementation of the Universal cost Function control Optimization (UFO) framework of Niu *et al.* [npj Quantum Inf. **5**, 33 (2019)] for the two-qubit gate family $N(\alpha,\alpha,\gamma) = \exp[i(\alpha\sigma^x\sigma^x + \alpha\sigma^y\sigma^y + \gamma\sigma^z\sigma^z)]$ on a three-level gmon model, and compare its trust-region policy optimization agent with a direct gradient optimizer that uses the same Hamiltonian, the same objective and the same training noise. Our time step is coarser than that of the original, which changes the noise scale but not the conclusions. The framework produces high-fidelity analog gates, but its central claims are not supported. Against per-step white control noise, the open-loop robustness of a pulse is fixed to leading order by its nominal fidelity and its duration, so training under such noise cannot make a pulse more robust. Matched pairs of agents and the direct optimizer confirm this, and the optimizer gains robustness only when it is trained against quasi-static noise. In our implementation the direct optimizer also beats the RL agents on every nominal metric. Finally, every member of the benchmark subfamily $N(\alpha,\alpha,\pi/2)$ is an exchange rotation by $\alpha - \pi/2$, so the reported order-of-magnitude runtime gain sits around the identity; against a target-specific two-CNOT synthesis the learned pulses are at most twice as fast.

I. INTRODUCTION

A quantum algorithm is written as a sequence of logical unitaries, but a quantum processor executes continuous-time dynamics generated by an experimentally accessible Hamiltonian. On pre-fault-tolerant superconducting hardware the translation between the two is not a formality: coherence budgets, calibration drift, waveform distortion and multilevel structure all limit which circuits can actually be run [1–3]. Quantum control is therefore not a fine-tuning layer on top of compilation, but one of the binding constraints on usable computational depth.

Rather than decomposing a target two-qubit unitary into a fixed universal gate alphabet and paying the runtime cost of every elementary operation, one may attempt to realize it directly through a single shaped analog pulse that exploits the native interactions of the device [4, 5]. If the target is well aligned with the hardware's own entangling generator, the analog implementation can be substantially shorter than any compiled circuit. There are two difficulties. Superconducting modes are weakly anharmonic, so fast or aggressively modulated pulses drive population out of the computational subspace [6, 7]; and the pulse that reaches the device is never the pulse that was designed, because amplitudes fluctuate and Hamiltonian parameters are known only approximately.

Niu *et al.* [4] proposed to attack these requirements simultaneously rather than in sequence. Their Universal cost Function control Optimization (UFO) framework combines three ingredients: an analytic leakage estimate derived from a time-dependent Schrieffer–Wolff transformation (TSWT), a scalar objective that adds leakage, boundary and runtime penalties to gate infidelity, and a continuous-action reinforcement-learning (RL) formulation trained with trust-region policy optimization (TRPO) [8] in an environment where control noise is injected during training. The reported outcome was strong: up to one order of magnitude lower average gate infidelity than an RL agent trained without noise, around two orders of magnitude lower than a stochastic gradient-descent (SGD) baseline, a fidelity standard deviation two orders of magnitude smaller than either, and a one-order-of-magnitude runtime reduction relative to optimal gate synthesis on the subfamily $N(\alpha,\alpha,\pi/2)$ for $\alpha \in [1.2, 1.7]$. The authors left one question explicitly open: whether the advantage of RL persists when gradients are cheap enough for gradient-based optimization to apply.

These are consequential claims, and they have been influential in establishing RL as a credible language for robust pulse design [9–11]. They have not, to our knowledge, been subjected to an independent reimplementation. That is the purpose of this work.

What this paper does. We rebuild the framework from scratch (multilevel gmon simulator, second-order TSWT leakage estimator, exact bandwidth filter, UFO objective, RL agent and stochastic environment) and use it to ask four questions. (i) Does the framework reproduce under an independent implementation, and with what quantitative fidelity? (ii) How does the RL agent compare with a direct pulse optimizer that computes exact gradients through the same simulator, with the same objective and the same training noise, the question left open by Ref. [4]? (iii) Does stochastic training make the deployed pulses more robust, and if not, why not? (iv) What does the runtime comparison with gate synthesis

actually measure, given the structure of the benchmark family? Along the way we test how much the conclusions depend on the free parameters the original paper does not pin down, the actor–critic architecture and the UFO weights.

What we find. The framework works as a pulse generator, but its central claims are not supported by controlled tests in our implementation. Items 1 and 3 below do not depend on the implementation; items 2 and 4 do.

1. *White noise leaves nothing to train for.* Against per-step white control noise, the open-loop robustness of any pulse is fixed to leading order by its nominal fidelity and its duration, Eq. (18), at a rate that we compute from the Hamiltonian and verify on every pulse we produced (Sec. V B). Accordingly, four matched pairs of RL agents that differ only in the training noise deploy pulses of the same robustness, and a direct optimizer gains robustness only when it is trained against quasi-static noise, by about a fifth (Secs. V D–V F).

2. *The direct optimizer is ahead.* On the representative target $N(2.2, 2.2, \pi/2)$ a single-start direct optimizer reaches about half the cost of our RL agents, with a pulse of 70 ns against 116 ns, and its pulse is the most robust at every noise strength (Secs. IV and V C). This compares one gradient optimizer with our agents; better-trained agents could narrow the gap.

3. *The benchmark family consists of exchange rotations.* Every member of $N(\alpha, \alpha, \pi/2)$ is an exchange rotation by $\alpha - \pi/2$: the target at $\alpha = \pi/2$ is the identity, the order-of-magnitude runtime gain of Ref. [4] lies around it, and every non-local member needs two CNOTs, not three (Sec. VI A). Against that synthesis the converged learned pulses are 1.0–2.0× faster.

4. *The runtime advantage belongs to analog control, not to learning.* The direct optimizer alone is 1.9–3.75× faster than two-CNOT synthesis, and its pulse durations track the exchange angle, close to the shortest time in which the bandwidth-limited coupling can deliver it (Sec. VI E).

Section II summarizes the framework and Sec. III our implementation; Secs. IV–VI present the nominal, robustness and runtime results; and Sec. VII compares them with the original claims. Derivations, the complete parameter set and secondary results are in the Supplemental Material [12].

II. CONTROL FRAMEWORK

This section summarizes the framework of Ref. [4].

A. Model and leakage bound

The platform is the gmon architecture: two weakly anharmonic superconducting modes with local drive and frequency control, coupled through a flux-tunable interaction [5], so that the entangling interaction is itself a control. In the rotating-wave approximation the two-mode Hamiltonian is [4]

$$\begin{aligned}\hat{H}_{\mathrm{RWA}}(t) = \frac{\eta}{2}\sum_{j=1}^{2}\hat{n}_j\big(\hat{n}_j - 1\big) + g(t)\big(\hat{a}_2^\dagger\hat{a}_1 + \hat{a}_1^\dagger\hat{a}_2\big) \\ + \sum_{j=1}^{2}\delta_j(t)\hat{n}_j \\ + \sum_{j=1}^{2} i f_j(t)\Big(\hat{a}_j e^{i\phi_j} - \hat{a}_j^\dagger e^{-i\phi_j}\Big),\end{aligned} \tag{1}$$

with $\hat{n}_j = \hat{a}_j^\dagger\hat{a}_j$, anharmonicity η, coupling g, detunings δ_j, and drives of amplitude f_j and phase ϕ_j: seven control channels. Projected onto the two lowest levels of each mode it becomes

$$\begin{aligned}\hat{H}_{\mathrm{qb}}(t) = \frac{g(t)}{2}\big(\sigma_1^x\sigma_2^x + \sigma_1^y\sigma_2^y\big) \\ + \sum_j\Big[\frac{\delta_j}{2}\sigma_j^z - f_j\big(\sin\phi_j\,\sigma_j^x + \cos\phi_j\,\sigma_j^y\big)\Big],\end{aligned} \tag{2}$$

an XY-type entangler with arbitrary single-qubit rotations, which is enough for universal two-qubit control. We use Eq. (2) for interpretation only: all dynamics are propagated with the multilevel Eq. (1), so that leakage is kept.

Because η is finite, the controls that produce the logical rotations also couple the computational sector $\Omega_0 = \mathrm{span}\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ to the leakage sector $\Omega_1 = \mathrm{span}\{|02\rangle, |20\rangle, |12\rangle, |21\rangle\}$. Decompose

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_1(t) + \hat{H}_2(t), \tag{3}$$

into the diagonal drift $\hat{H}_0$, a block-diagonal part $\hat{H}_1$ and a block-off-diagonal part $\hat{H}_2$, with Δ the minimum gap between Ω_0 and the leakage sectors and $\epsilon \sim \|\hat{H}_1\| \sim \|\hat{H}_2\| \ll \Delta$ the control scale. A time-dependent Schrieffer–Wolff transformation (TSWT) moves to a frame $|\tilde{\psi}\rangle = e^{-\hat{S}(t)}|\psi\rangle$, with anti-Hermitian $\hat{S}$, in which the Hamiltonian is

$$\hat{\mathbb{H}}(t) = -i\,\frac{de^{-\hat{S}}}{dt}e^{\hat{S}} + e^{-\hat{S}}\hat{H}e^{\hat{S}}, \tag{4}$$

and chooses $\hat{S}$ to cancel the block-off-diagonal terms order by order. At second order a residual $\hat{H}_{\mathrm{od}} = O(\epsilon^3/\Delta^2)$ is left, and it bounds the coherent leakage accumulated over a gate of duration T (Supplemental Material [12],

Sec. S1):

$$L_{\rm tot}(T) \lesssim \frac{\|\hat{H}_{\rm od}(0)\|}{\Delta(0)} + \frac{\|\hat{H}_{\rm od}(T)\|}{\Delta(T)} + \int_0^T \frac{1}{\Delta(t)^2} \left\| \frac{d^2 \hat{H}_{\rm od}(t)}{dt^2} \right\| dt. \quad (5)$$

The two boundary terms penalize residual off-diagonal coupling at the pulse ends. The integral penalizes rapid modulation: a pulse can be small in amplitude at every instant and still leak if it changes too quickly. Because the bound is written in terms of the Hamiltonian and its derivatives, it can be accumulated during propagation.

B. The UFO objective

Let $U(T) = \mathcal{T}\exp[-i\int_0^T \hat{H}_{\rm RWA}(t)dt]$ and let $U_{\rm proj}$ be its restriction to Ω_0. The gate fidelity with respect to $U_{\rm target}$ is

$$F = \frac{1}{16}\left|\mathrm{Tr}\left(U_{\rm proj}^\dagger(T)\, U_{\rm target}\right)\right|^2, \quad (6)$$

which equals 1 when the implemented gate matches the target up to a global phase. The objective adds three penalties [4],

$$C(\chi,\beta,\mu,\kappa) = \chi\left[1-F\right] + \beta L_{\rm tot} + \kappa T + \mu \sum_{t\in\{0,T\}} \left[g(t)^2 + f_1(t)^2 + f_2(t)^2\right]. \quad (7)$$

for leakage, for runtime and for control envelopes that do not vanish at the pulse edges. All four terms are soft constraints: every control channel stays available, and the optimizer trades the objectives against one another.

C. Reinforcement-learning formulation

The pulse is built step by step. Time is discretized in steps Δt; the state presented to the agent is the propagator accumulated so far, $s_n \sim U_n$; the action is the control vector for the next interval; and the environment propagates

$$U_{n+1} = \exp\left[-i\Delta t(\hat{H}_{n+1} + \delta\hat{H}_{n+1})\right]U_n, \quad (8)$$

where $\delta\hat{H}$ is the stochastic perturbation used in noisy training. The reward is the negative running UFO cost, $r_n = -c_n$. An episode ends at $T_{\max}$ or at the first time the cost falls below a threshold, so the agent optimizes the stopping time together with the pulse shape, unlike a fixed-horizon pulse optimizer. The policy is trained with TRPO [8], which maximizes the importance-weighted surrogate

$$\mathcal{L}(\boldsymbol{\theta}) = \mathbb{E}_{\pi_{\boldsymbol{\theta}_{\rm old}}}\left[\frac{\pi_{\boldsymbol{\theta}}(a|s)}{\pi_{\boldsymbol{\theta}_{\rm old}}(a|s)}\hat{A}(s,a)\right] \quad \text{s.t.} \quad \mathbb{E}_s\left[D_{\rm KL}\right] \le \delta_{\rm KL}. \quad (9)$$

where the constraint limits how far one update can move the policy.

In stochastic training, Gaussian fluctuations are added to the control amplitudes at every step, so that the objective becomes an expectation over both the policy and the noise, $J(\boldsymbol{\theta}) = -\mathbb{E}_{\tau,\xi}[C]$, and pulse trajectories that are fragile under small perturbations earn a poor average return. We call *noise-free* the agent trained in the deterministic environment and *noise-trained* the one trained in the stochastic environment (the "noise-optimized" controller of Ref. [4]). They differ only in the training distribution.

Two details of this construction matter later. First, the agent observes the propagator produced by the *noisy* dynamics, so during training it can react to the realized noise; the pulse that is deployed and evaluated is the open-loop pulse of its mean action in the noise-free environment, which cannot. Second, every term of Eq. (7) is evaluated on the noisy trajectory, including the leakage bound, whose time derivatives respond to the noise itself (Appendix B).

D. Target family and synthesis reference

Benchmarking uses the family

$$N(\alpha,\alpha,\gamma) = \exp\left[i\left(\alpha\sigma_1^x\sigma_2^x + \alpha\sigma_1^y\sigma_2^y + \gamma\sigma_1^z\sigma_2^z\right)\right]. \quad (10)$$

The equal XX and YY coefficients mirror the exchange structure of the gmon coupling. Up to single-qubit rotations the family contains SWAP, iSWAP, CNOT and CZ, as well as the fermionic SWAP and Givens rotations of Jordan–Wigner-mapped fermionic simulation [4].

The runtime reference of Ref. [4] is a generic circuit decomposition: three CNOTs and four layers of single-qubit rotations, the Vatan–Williams construction, which is optimal for an *arbitrary* two-qubit unitary [13]. With the gate times assumed there, $T_{1q} = 20\,\mathrm{ns}$ and $T_{\rm CNOT} = 45\,\mathrm{ns}$, it takes

$$T_{\rm syn} \approx 215\,\mathrm{ns}. \quad (11)$$

We keep this number for comparability, but it is not the optimal synthesis of each member of the family: for the subfamily $\gamma = \pi/2$ studied here two CNOTs suffice, which take 150 ns [Eq. (20)], and three members need no entangling gate at all (Sec. VI A).

III. NUMERICAL IMPLEMENTATION

A. Simulator

Each mode is truncated to three levels, giving $d_{\rm phys} = 9$ and $d_{\rm comp} = 4$, and one-step propagation uses exact diagonalization of the Hermitian generator. The anharmonicity is $\eta = -200\,\mathrm{MHz}$, which also sets the gap scale entering Eq. (5). Control amplitudes are bounded at 20 MHz,

well below $|\eta|$ so that the perturbative regime $\epsilon/\Delta \ll 1$ is respected, and phases are mapped to $[0, 2\pi]$.

Before reaching the Hamiltonian, the piecewise-constant controls pass through the two-pole double-exponential filter of Ref. [4], implemented exactly: with $a = \exp(-\pi B/f_s)$, $B = 10\,\mathrm{MHz}$ the bandwidth and $f_s = 1/\Delta t$ the sampling rate,

$$\tilde{u}_k = (1-a)^2 u_k + 2a\,\tilde{u}_{k-1} - a^2\,\tilde{u}_{k-2}. \tag{12}$$

The filter keeps the optimizer from exploiting control transitions sharper than a waveform generator can produce, and it interacts directly with the non-adiabatic term of the leakage bound.

The leakage bound is evaluated numerically at every step. The estimator solves the second-order TSWT for the generators S_1 and S_2, takes their time derivatives by finite differences, forms the residual $H_{\mathrm{od}}^{(3)}$ and accumulates the boundary and integral terms of Eq. (5) incrementally, at a cost per step that does not grow with the pulse length (Supplemental Material [12], Sec. S2).

B. Noise model

At each step the realized control is $\boldsymbol{u}^{(\mathrm{noisy})}(t_k) = \boldsymbol{u}(t_k) + \boldsymbol{\xi}(t_k)$ with $\boldsymbol{\xi}$ drawn independently from a zero-mean Gaussian of standard deviation σ_{noise}. Perturbations are applied to the coupling g, the detunings $\delta_{1,2}$ and the drive amplitudes $f_{1,2}$, and additionally to the anharmonicity η; phases are left noise-free. Including η is a deliberate feature of the model: it probes sensitivity not only to imperfect pulse delivery but also to uncertainty in a device parameter that no control can actuate. The noise is added after the bandwidth filter and is independent from one 2 ns step to the next, so it is white up to the 250 MHz Nyquist frequency of the grid, above the 200 MHz gap to the leakage sector. Ref. [4] discretizes its pulses more finely, in "around one thousand time steps"; the filter time constant of 32 ns is well resolved at 2 ns, so the noise-free control problem is essentially the same.

The step does change the meaning of σ_{noise}, and this is a caveat for every comparison with Ref. [4]. Because the noise is redrawn at every step, the same σ_{noise} is a stronger physical noise at a coarser step: at our 2 ns it costs about sixteen times more infidelity than at a step of 0.1 ns. The σ_{noise} axes of the two studies should therefore not be identified, and our fidelities under noise are not to be compared with those of Ref. [4] at equal σ_{noise} (Sec. VII B and Appendix C). None of our conclusions depends on this scale.

Training noise and evaluation noise are kept separate. The noise-free agent is optimized without perturbations. The noise-trained agent and the direct optimizer are both optimized under this perturbation model at $\sigma_{\mathrm{train}} = 1\,\mathrm{MHz}$ (Sec. III C). All pulses are then probed *a posteriori* on the evaluation grid of Sec. V.

C. Methods compared

Direct optimizer. For a fixed target and a fixed horizon T, the full discretized pulse $\{g_\ell, \delta_{j,\ell}, f_{j,\ell}, \phi_{j,\ell}\}_{\ell=0}^{N_T-1}$ is optimized directly with Adam [14] on the same multilevel model, the same filter and the same objective Eq. (7), differentiating exactly through the simulation: a GRAPE-style direct pulse search [3, 15] rather than a literal reproduction of the canonical algorithm. It is trained under the same noise model as the noise-trained agent: at every step the loss is the cost of one trajectory perturbed at $\sigma_{\mathrm{train}} = 1\,\mathrm{MHz}$, a single-sample stochastic gradient of $\mathbb{E}_\xi[C]$. Since the horizon is not a free variable of a single run, the procedure is repeated over $T \in \{40, 50, \ldots, 180\}$ ns, with 400 iterations per horizon, learning rate 0.03 and a single deterministic initialization (all amplitudes zero, phases π): 3.3×10^5 simulator steps in all, against 9.5×10^7 for each RL agent, a ratio of about 290. Because single-sample losses are noisy, every horizon's pulse is re-evaluated in the deterministic simulator, and the pulse we report for the single-target benchmark is the one with the lowest deterministic cost.

By the standards of Ref. [4] this baseline is modest. The SGD baseline there also used Adam and a noisy objective, but averaged ten noisy trajectories per gradient step and received about 2000 random restarts per target. What differs here is that the gradients are exact and are computed through the same filtered multilevel simulator in which the RL agents are trained. The comparison therefore addresses the question Ref. [4] left open: whether the advantage of reinforcement learning persists when gradient estimation is cheap enough for gradient-based optimization to apply, rather than whether a weak baseline can be beaten. The baseline is a family of fixed-horizon problems, whereas an RL agent learns a variable-time policy; this structural difference is part of what the comparison tests.

RL agents. Both agents are trained with TRPO and use the same actor–critic pair. The observation vectorizes the current 9×9 propagator into real and imaginary parts and appends the normalized time index,

$$s_t = \big(\Re\,\mathrm{vec}(U_t),\ \Im\,\mathrm{vec}(U_t),\ t/T_{\max}\big), \qquad d_{\mathrm{obs}} = 163. \tag{13}$$

Actor and critic are multilayer perceptrons with hidden sizes $(64, 32, 32)$ and tanh activations, as in Ref. [4]. The actor emits a mean $\mu_\theta(s_t) \in \mathbb{R}^7$ with a state-independent trainable $\log\sigma$; raw actions $\tilde{a}_t = \mu_\theta + \sigma \odot \zeta_t$, with ζ_t standard normal, are squashed by tanh into $[-1, 1]^7$ and rescaled to physical bounds. The critic outputs a scalar baseline for advantage estimation. At every iteration the mean-action rollout in the noise-free environment is stored as a pulse, and the pulse reported for an agent is the one with the lowest nominal cost over the whole training run.

Gate synthesis. The generic compiled reference of Eq. (11) and the target-specific one of Eq. (20), used for runtime only.

D. Metrics

Nominal quality is reported through the gate fidelity Eq. (6), the TSWT leakage estimate L_{tot} with its boundary/integral split, the boundary penalty, the effective runtime and the total cost Eq. (7).

Robustness is characterized by two distinct quantities. The first is the average fidelity of the *noise-averaged channel* $\mathcal{E}$, estimated from the Monte Carlo ensemble of noisy projected unitaries $\{K_j\}$ through the standard formula [16]

$$\overline{F} = \frac{\sum_j \mathrm{Tr}\big(U_{\mathrm{target}} U_j^\dagger U_{\mathrm{target}}^\dagger \mathcal{E}(U_j)\big) + d^2}{d^2(d+1)}, \qquad d = 4, \tag{14}$$

with $\{U_j\}$ a two-qubit Pauli basis. The second is the variance across realizations of the per-shot gate fidelity,

$$\mathrm{Var}(F) = \mathbb{E}_\xi\big[(F_\xi - \mathbb{E}_\xi[F_\xi])^2\big]. \tag{15}$$

A pulse with good mean fidelity but large variance is experimentally fragile, so neither quantity alone settles the question.

E. Parameters

All runs share the physical model above, a time step $\Delta t = 2\,\mathrm{ns}$ and a maximum gate duration $T_{\max} = 180\,\mathrm{ns}$; the runtime in Eq. (7) is measured in units of $T_{\mathrm{norm}} = 60\,\mathrm{ns}$. The complete configuration of every experiment is listed in the Supplemental Material [12], Sec. S3.

Two ingredients of the framework are not determined by the physics: the capacity of the actor–critic networks and the relative weights in Eq. (7). We fixed them first, with two sweeps on the representative target $N(2.2, 2.2, \pi/2)$ under a reduced budget (Supplemental Material [12], Sec. S4). Neither sweep gives a reason to depart from the published choices: the layout $(64, 32, 32)$ and the weights

$$(\chi, \beta, \mu, \kappa) = (10, 10, 0.2, 0.1) \tag{16}$$

are within the seed-to-seed spread of the best configurations we found, and we use them throughout. The sweeps also give that spread: three seeds of the same configuration reach $\overline{F} = 0.959$, 0.962 and 0.965 at $\sigma_{\mathrm{noise}} = 1\,\mathrm{MHz}$. Differences of a few 10^{-3} in $\overline{F}$ between two individual runs are therefore not significant on their own, which sets the resolution of every single-seed comparison below.

IV. SINGLE-TARGET BENCHMARK

All methods are first compared on

$$U_{\mathrm{target}} = N(2.2, 2.2, \pi/2), \tag{17}$$

a target that is neither near-identity nor a degenerate special case and that sits in the family used for the original robustness analysis. Table I collects the three pulses.

TABLE I. Nominal performance on $N(2.2, 2.2, \pi/2)$: duration, cost, fidelity and leakage bound of each pulse, evaluated without noise; best value in each column in bold. The synthesis rows are the generic reference of Ref. [4], Eq. (11), and the two-CNOT reference of Eq. (20), shown for runtime only.

Method	T (ns)	C^{nom}	F^{nom}	L_{tot}
Direct optimizer	**70**	**0.1576**	**0.99662**	$\mathbf{2.82 \times 10^{-4}}$
RL, noise-free	116	0.2982	0.99415	1.60×10^{-3}
RL, noise-trained	98	0.3537	0.99078	4.73×10^{-4}
Generic synthesis	215	—	—	—
Two-CNOT synthesis	150	—	—	—

The direct optimizer finds its best pulse at a horizon of 70 ns, with fidelity 0.9966 and leakage in the 10^{-4} range. Shorter pulses fail because they cannot be both accurate and zero at their ends, and longer ones pay in runtime without gaining accuracy (Supplemental Material [12], Sec. S5). Optimized without training noise at the same horizon the pulse is better still, with fidelity 0.99939 (Sec. V F).

Both RL agents also produce high-quality pulses. The noise-free agent reaches fidelity 0.994 in 116 ns, and the noise-trained agent 0.991 in 98 ns.[1] Both stop well before the 180 ns ceiling, so the agents do optimize the duration together with the shape, and both are faster than the two synthesis references.

The direct optimizer is nevertheless ahead of both agents on all four metrics: against the noise-free agent by 1.9× in cost, 1.7× in runtime and 5.7× in leakage. That is what one should expect: when the model is known exactly and the objective is differentiable, direct gradient methods are very effective on fixed-target closed-system pulse design [3, 15]. The baseline against which the robustness claims are tested is therefore a capable one, even though its budget is small. The noise-trained agent's pulse is shorter and less leaky than the noise-free agent's, but less accurate and more expensive: stochastic training has not bought nominal quality. Whether it has bought robustness is the subject of the next section.

[1] This is the pulse with the lowest nominal cost over the whole run, from iteration 11, the rule applied to the noise-free agent. The pulse that the training script saved as its best, from iteration 91 (104 ns, fidelity 0.988), comes from a resumed segment of the run that restarted the script's best-tracking; it is evaluated in Sec. V C as well.

V. ROBUSTNESS UNDER STOCHASTIC CONTROL NOISE

A. Evaluation protocol

Each pulse is replayed open loop under the noise model of Sec. III B on a dense grid of 3401 noise strengths, $\sigma_{\text{noise}} \in [0.1, 3.5]$ MHz in steps of 0.001 MHz, with 60 independent realizations at each point. Sixty samples give a noisy point estimate, so the curves are read through an exponentially weighted moving average along the σ_{noise} axis (span 50 for the mean, 80 for the variance). The underlying scatter is shown in the figures and kept in the released data, and no other post-processing is applied. Every pulse is scored against $N(2.2, 2.2, \pi/2)$, with the same seed at each grid point.

We compare the three pulses of Table I, together with a shorter, 60 ns pulse from the same direct optimizer (fidelity 0.982; it is the one that the optimizer's noisy objective selects, Appendix B). A few individual pulses cannot isolate the effect of stochastic training, since each is one draw from a training process. We therefore first state what the noise model allows any pulse to gain (Sec. V B), and then test it on the selected pulses (Sec. V C), on the whole training histories of matched agents (Sec. V D) and with the direct optimizer (Sec. V F).

B. What white noise allows

Under this noise model the pulse shape cannot buy robustness. This is the white-noise limit of the filter-function description of open-loop control [17–19]: the response of a pulse to noise is the overlap of the noise spectrum with a filter function fixed by the pulse, and a flat spectrum sees only the integral of that function, which the pulse shape does not change (Appendix A).

In the present model the argument is short. The noise is independent from one 2 ns step to the next. Moving a kick to the end of the pulse conjugates its generator G by the evolution so far, and as long as that evolution keeps the computational subspace invariant, the first-order infidelity of the kick depends only on the variance $V(G) = \mathrm{Tr}(PG^2P)/d - |\mathrm{Tr}(PGP)/d|^2$ of G over that subspace P, which conjugation leaves unchanged. To leading order the noise-induced infidelity is therefore independent of the pulse shape and proportional to the duration,

$$1 - \overline{F}(\sigma_{\text{noise}}) \simeq 1 - \overline{F}(0) + K\sigma_{\text{noise}}^2 T,$$
$$K = \frac{d}{d+1}\left(2\pi \cdot 10^{-3}\right)^2 \Delta t \sum_c V_c, \qquad (18)$$

with σ_{noise} in MHz, Δt and T in ns, and the sum running over the noisy channels. The components of G that lead out of the subspace, to states detuned by the anharmonicity, are partly averaged during the step; we weight them by the zero-order-hold factor $\mathrm{sinc}^2(\omega\Delta t/2)$ at their transition frequency ω. For the gmon model this gives $V = 1/4$ for each detuning, 1.57 for each drive, 1.07 for the coupling and 0 for the anharmonicity, and hence $K = 2.98 \times 10^{-4}\,\mathrm{ns}^{-1}\,\mathrm{MHz}^{-2}$. Two thirds of it come from the two drive amplitudes, and replaying the pulses with the noise on one channel at a time reproduces each channel's share (Table VI).

The prediction holds on every pulse we produced. For the 160 pulses stored along the training of a matched pair of RL agents (Sec. V D), the measured excess infidelity per unit duration at 1 MHz is 2.97×10^{-4} and 2.99×10^{-4} for the noise-free and the noise-trained agent, with interquartile ranges within ±2%. Evaluated with the exact response of every step instead of the sinc approximation (Appendix A), the first-order rate of those pulses varies by four parts in 10^4: that is all the pulse shape can move. Excess infidelity and duration are correlated at 0.99 across both agents, and the pulses of the direct optimizer fall on the same line [Fig. 2(a)]. Away from 1 MHz the measured rate drifts, to 3.4×10^{-4} at 0.5 MHz and 2.7×10^{-4} at 2 MHz, which reflects higher orders and the approximate treatment of the leaking components.

Against white control noise an open-loop pulse is robust only by being accurate and short, and a noise-aware objective can improve robustness only through those two numbers. The rest of this section tests that statement.

C. Selected pulses

Figure 1 and Table II give the result for the selected pulses. Among the three pulses of Table I, that of the direct optimizer has the highest average fidelity and the lowest variance at every noise strength. At the training noise it reaches $\overline{F} = 0.9782$, against 0.9659 for the noise-trained agent and 0.9631 for the noise-free one.

Between the two RL pulses the order depends on the noise strength. Below $\sigma_{\text{noise}} \approx 0.72$ MHz the noise-free agent's pulse is better, as its higher nominal fidelity dictates; above, the noise-trained one is, with an infidelity ratio of 1.08 at 1 MHz that saturates near 1.14 at large noise. That is an order of magnitude below the gain reported in Ref. [4], and the order follows the durations, as Eq. (18) requires. At 3.5 MHz the four pulses of Fig. 1 and the 104 ns pulse of the noise-trained agent (Sec. IV; not shown) rank exactly by duration (60, 70, 98, 104 and 116 ns), whichever method produced them, and the 60 ns pulse, although less accurate, overtakes the 70 ns one above about 2 MHz. Section V D shows that even the difference between the two RL pulses does not come from stochastic training.

D. Matched agents

Two selected pulses cannot isolate the effect of the training noise, but a *matched pair* of agents can: two RL agents trained from the same seed, with the same

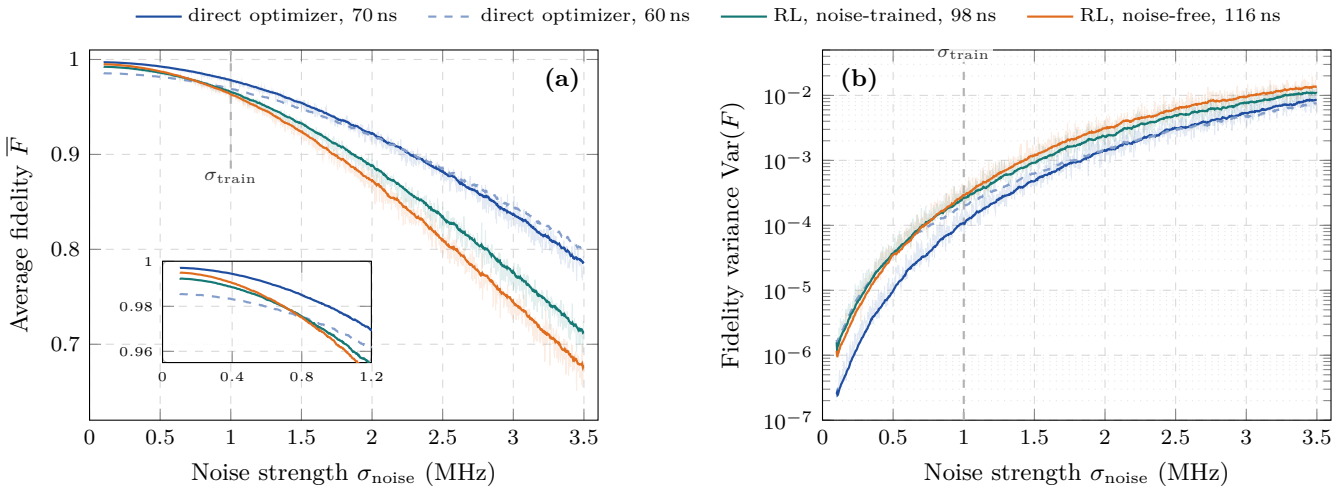


FIG. 1. Robustness on $N(2.2, 2.2, \pi/2)$ of the pulses of Table I and of a shorter, 60 ns pulse from the direct optimizer, replayed open loop. (a) Average fidelity $\overline{F}(\sigma_{\text{noise}})$, with the low-noise regime magnified in the inset. (b) Fidelity variance on a logarithmic axis. Lines are exponentially weighted moving averages along the noise axis (spans 50 and 80); the underlying Monte Carlo scatter is shown for three of the curves. The dashed vertical lines mark the training noise $\sigma_{\text{train}} = 1$ MHz. The 70 ns pulse of the direct optimizer is more robust than both RL pulses at every noise strength, and at high noise the four pulses rank by duration.

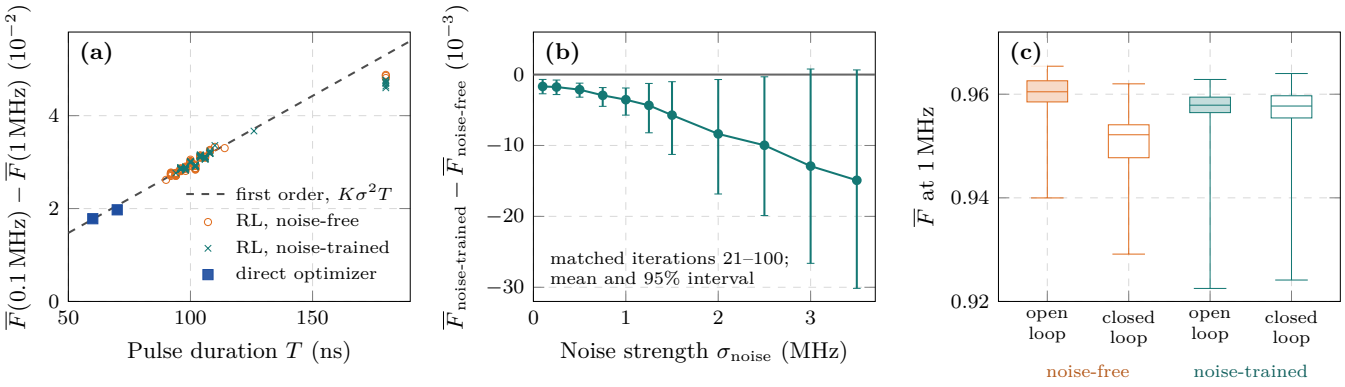


FIG. 2. Open-loop robustness and stochastic training. (a) Excess infidelity at 1 MHz against pulse duration for all pulses of a matched pair of RL agents and for the pulses of the direct optimizer, with the first-order prediction of Eq. (18) (dashed); the pulses at 180 ns are rollouts that reached the episode ceiling, where higher orders begin to matter. (b) Paired difference in average fidelity between the noise-trained and the noise-free pulse of the same iteration, for the first matched pair over iterations 21–100, with 95% block-bootstrap intervals; negative values favour the noise-free agent. (c) Average fidelity at 1 MHz of the same checkpoints, replayed open loop (filled) and acting in closed loop on the noisy propagator (open); boxes span the quartiles and whiskers the full range.

architecture, weights, batch size and stopping cost, for 100 iterations, one without noise and one with it. At every iteration each agent stores the pulse of its current policy, and comparing those pulses iteration by iteration isolates the training noise much better than comparing two selected pulses.

We trained four such pairs. The first is the pair of Table I. In the other three, with seeds 1, 2 and 3, the noise-trained agent accumulates the leakage bound from the noise-free controls, so that the training noise enters its objective through the fidelity alone; Appendix B explains why this matters.[2] For each pair we evaluated the 160 pulses of iterations 21–100 at 11 noise strengths between 0.1 and 3.5 MHz, with 200 realizations each.

The result is the opposite of the intended effect [Table III and Fig. 2(b)]. In every pair the noise-trained

[2] These three pairs use a batched implementation of the environment, which agrees with the original one to 10^{-13} in the cost at every step but draws different random numbers, so its seed 1 is not the first pair.

TABLE II. Robustness at three noise strengths (σ_{noise} in MHz) of the pulses of Table I. Upper block: channel average fidelity $\overline{F}$, Eq. (14). Lower block: fidelity variance. All values are EWMA estimates over the dense grid of Fig. 1; the best pulse at each noise strength is in bold.

σ_{noise}	Direct 70 ns	RL, noise-free 116 ns	RL, noise-trained 98 ns
0.1	**0.9970**	0.9950	0.9925
1.0	**0.9782**	0.9631	0.9659
3.5	**0.7848**	0.6764	0.7128
Fidelity variance Var(F)			
0.1	$\mathbf{2.5\times10^{-7}}$	1.2×10^{-6}	1.6×10^{-6}
1.0	$\mathbf{1.1\times10^{-4}}$	2.9×10^{-4}	2.7×10^{-4}
3.5	$\mathbf{8.5\times10^{-3}}$	1.4×10^{-2}	1.1×10^{-2}

TABLE III. Matched comparison at $\sigma_{\text{noise}} = 1$ MHz, iterations 21–100. Pair 1 is the pair of Table I; in pairs 2–4 (seeds 1–3) the leakage bound of the noise-trained agent is taken from the noise-free controls. Rows: fraction of iterations in which the noise-trained pulse has the higher $\overline{F}$; mean paired difference in $\overline{F}$ (noise-trained minus noise-free); mean difference in duration; and the difference in $\overline{F}$ that Eq. (18) predicts from the durations and nominal fidelities.

	Pair 1	Pair 2	Pair 3	Pair 4
Noise-trained better	21%	26%	28%	24%
$\Delta\overline{F}$ (10^{-3})	−3.5	−1.6	−1.7	−2.0
ΔT (ns)	—	+5.1	+5.7	+5.2
Predicted $\Delta\overline{F}$ (10^{-3})	—	−1.9	−1.6	−2.2

pulse is *less* robust than the noise-free pulse of the same iteration in most iterations: at 1 MHz it has the higher average fidelity in only 21–28% of them. The mean paired difference at 1 MHz is between −1.6 and -3.5×10^{-3}, and its 95% block-bootstrap interval excludes zero in all four pairs.

Two remarks bound this result. First, the difference is of the size of the seed-to-seed spread (Sec. III E): pairing each noise-trained agent of pairs 2–4 with each noise-free one, whatever their seeds, gives differences between −3.5 and $+0.2\times10^{-3}$ at 1 MHz. What the four pairs establish is therefore the absence of a gain, not a loss intrinsic to stochastic training. Second, the difference is not a matter of pulse shape: the noise-trained pulses are 5–6 ns longer on average and differ in nominal fidelity by 10^{-3} or less, and Eq. (18) turns those two numbers into the difference measured (Table III).

What the data do rule out is an improvement of the size reported in Ref. [4]. An order of magnitude in infidelity at 1 MHz would mean $\overline{F} \approx 0.996$ instead of 0.96, far outside the spread of every agent: no pulse of pairs 2–4 exceeds $\overline{F} = 0.968$ at 1 MHz.

E. What stochastic training changes instead

What stochastic training does change is the policy's response to noisy states, and that response is not deployed. During training the agent observes the noisy propagator and acts on it. We ran every checkpoint of the first matched pair in the noisy environment at 1 MHz, in closed loop and for the duration of its own pulse, and compared it with the open-loop replay of that pulse [Fig. 2(c)]. In closed loop the noise-trained agent *is* the more robust one: it beats the noise-free agent in 81% of iterations, by 5.4×10^{-3} in $\overline{F}$ on average (95% interval $[3.4, 7.4]\times10^{-3}$). The reason is that the noise-free agent, fed states it never saw in training, reacts badly: its closed-loop fidelity falls below its own open-loop value in 99% of iterations, by 8.8×10^{-3} on average, with three times the variance. Stochastic training removes this instability (the noise-trained agent's closed-loop and open-loop fidelities agree within 10^{-3}) but it neither produces feedback that beats open loop nor changes the open-loop pulse. A robustness test carried out in the training environment would credit the noise-trained agent with an advantage that its deployable pulses do not have.

In our implementation stochastic training also changes the objective itself. The leakage bound, evaluated on the noisy trajectory, responds to the noise and acts as a strong implicit runtime penalty. This effect depends on the time step and on where the bound is evaluated, and none of the results above rests on it; Appendix B gives the details.

F. Noise with memory, and the same test with the direct optimizer

Equation (18) makes two predictions that can be tested without retraining an agent. The first is that the pulse shape should matter as soon as the noise is correlated in time; the second, that a noise-aware optimizer should then be able to use it, and should gain nothing under white noise.

Existing pulses. We replayed every stored pulse under quasi-static noise, one Gaussian offset per channel held for the whole pulse and redrawn for every shot, and evaluated Eq. (A2) for it (Appendix A; the two agree within 3% at 0.1 MHz). At equal σ_{noise} the infidelity is 15 to 23 times the white-noise one for the pulses of Table I, because the offsets of successive steps add coherently, and it grows as T^2 instead of T [Fig. 3(c)], so short pulses gain even more than under white noise. The shape now enters, but among pulses that were not optimized for it, only weakly: divided by T^2, the sensitivity of the 160 pulses of the first matched pair spans 5.7 to 6.3×10^{-7} ns^{-2}, and the noise-trained agent is not ahead. It has the higher average fidelity in 25% of the matched iterations at 0.1 MHz and 20% at 0.3 MHz. Training under white noise has not prepared the policy for noise with memory either.

The direct optimizer under three noise models. We then repeated the comparison of Sec. V D with the gradient optimizer, where it costs minutes. At a fixed horizon, so that the duration cannot change, the pulse was

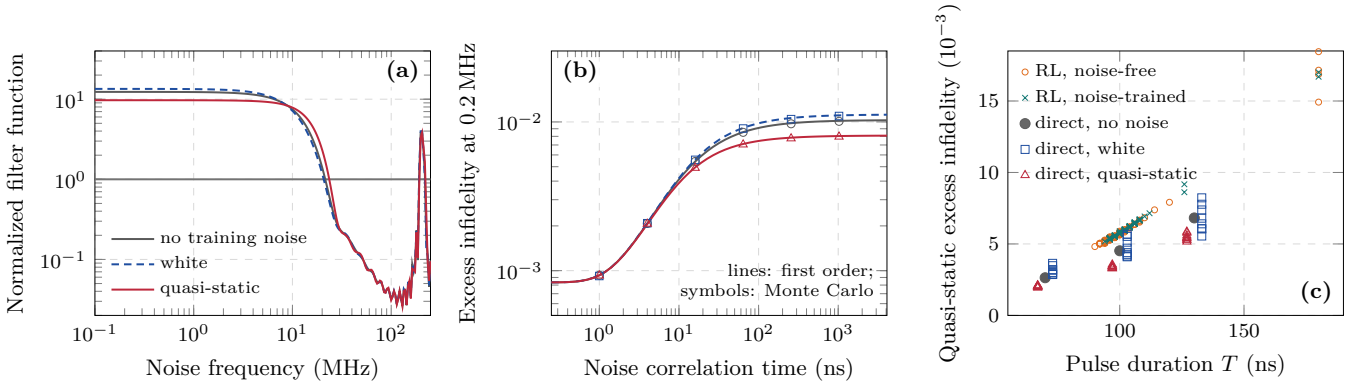


FIG. 3. Noise with memory. (a) Filter functions, summed over the channels and normalized to their average over the Nyquist band, of three 70 ns pulses of the direct optimizer for $N(2.2, 2.2, \pi/2)$: optimized without noise, under white noise (1 MHz) and under quasi-static noise (0.3 MHz). The average is the same for the three (horizontal line); the low-frequency plateau is not. The peak at 200 MHz is the transition to the leakage levels. (b) Excess infidelity of the same pulses under exponentially correlated noise of standard deviation 0.2 MHz against its correlation time: first-order prediction of Eq. (A2) (lines) and Monte Carlo replay (8000 realizations, symbols). (c) Excess infidelity under quasi-static noise of 0.1 MHz against duration, for the pulses of the first matched pair of RL agents and for pulses of the direct optimizer optimized at 70, 100 and 130 ns under the three noise models (eight seeds for each stochastic model; the white and quasi-static symbols are shifted by ±3 ns for visibility).

TABLE IV. The direct optimizer at fixed horizon T under three training-noise models (mean over eight seeds for the stochastic ones; the noise-free optimization is deterministic). $1-F$: nominal infidelity. White: average fidelity $\overline{F}$ at 1 MHz from 2000 realizations. Quasi-static: first-order excess infidelity at 0.1 MHz, Eq. (A2), and $\overline{F}$ at the training strength of 0.3 MHz.

			White	Quasi-static	
T	Training	$1-F$	$\overline{F}$	excess	$\overline{F}$
(ns)	noise	(10^{-3})	1 MHz	(10^{-3})	0.3 MHz
70	none	**0.6**	**0.9792**	2.57	0.9762
	white	5.4	0.9757	3.04	0.9682
	quasi-static	3.5	0.9767	**2.01**	**0.9792**
100	none	**0.3**	**0.9707**	4.43	0.9602
	white	5.4	0.9667	4.73	0.9541
	quasi-static	5.9	0.9663	**3.38**	**0.9654**
130	none	**0.2**	**0.9624**	6.71	**0.9411**
	white	10.3	0.9546	6.75	0.9334
	quasi-static	16.4	0.9504	**5.47**	0.9393

optimized (i) on the deterministic objective, (ii) under the white training noise of the paper, $\sigma_{\text{train}} = 1$ MHz, and (iii) under quasi-static noise of 0.3 MHz, one offset per channel and per trajectory, a single-sample version of ensemble-based robust control [20]; eight seeds for each stochastic model, horizons of 70, 100 and 130 ns, otherwise the settings of Sec. III C. Table IV gives the result.

Under white noise the pulses of a given horizon have the same first-order excess infidelity whatever the training noise, to better than one part in a thousand (2.09, 2.98 and 3.87×10^{-2} at 1 MHz for the three horizons), as Eq. (18) requires, and the only thing that white-noise training changes is the nominal fidelity, for the worse: none of the 24 white-trained pulses is more accurate, or more robust against the very noise it was trained on, than the pulse optimized without noise at the same horizon. Its stochastic gradient costs 5–10 $\times 10^{-3}$ in nominal fidelity and buys nothing. The null result of Sec. V D is therefore not a peculiarity of the RL agents: it recurs, with seeds, in an optimizer whose gradients are exact.

Part of that cost comes from the leakage bound, which in our objective responds to the noise itself. With the bound taken from the noise-free controls the cost is smaller and the outcome is the same: over the two variants, white-noise training helped in none of 48 runs (Appendix B).

Quasi-static training does what white training cannot. It lowers the quasi-static sensitivity in every one of the 24 runs, by 18–24% on average and with a spread of a few percent between seeds, by moving weight of the filter function away from zero frequency while leaving its average untouched [Fig. 3(a)]; the three pulses coincide for correlation times below a few nanoseconds and separate beyond [Fig. 3(b)]. The gain is modest, because a 70–130 ns pulse through a 10 MHz filter has little room for the refocusing structure of dynamically corrected gates [21], and it is paid for in nominal fidelity, so it shows in $\overline{F}$ only once the noise is strong enough: at 0.1 MHz a single quasi-static-trained pulse beats the noise-free one, at the training strength of 0.3 MHz 19 of 24 do, against 2 of 24 white-trained ones.

Robustness that depends on the pulse shape therefore requires noise with memory (like quasi-static offsets, drift or low-frequency spectra) which is what robust-control techniques are designed for [3] and what the UFO noise model does not contain.

VI. RUNTIME ACROSS THE GATE FAMILY

A. Structure of the subfamily $\gamma = \pi/2$

The runtime study uses $\gamma = \pi/2$, and there the two-qubit structure can be stated exactly. Since $\exp(i\frac{\pi}{2}\sigma_1^z\sigma_2^z) = i\sigma_1^z\sigma_2^z$ and $\exp[i\frac{\pi}{2}(\sigma_1^x\sigma_2^x+\sigma_1^y\sigma_2^y)] = \sigma_1^z\sigma_2^z$, every member is, up to a global phase, a pure exchange rotation,

$$N(\alpha,\alpha,\pi/2) = i\exp\big[i\varepsilon\big(\sigma_1^x\sigma_2^x+\sigma_1^y\sigma_2^y\big)\big], \qquad \varepsilon = \alpha - \frac{\pi}{2}, \tag{19}$$

which is the decomposition noted in Ref. [4] with its local Z rotations absorbed. Four consequences are used below.

(i) *Local members.* At $\alpha = \pi/2$ the target is the identity, and at $\alpha = 0$ and $\alpha = \pi$ it is the local gate $\sigma_1^z\sigma_2^z$. Because the fidelity (6) ignores global phases, a pulse that does nothing implements the $\alpha = \pi/2$ target exactly.

(ii) *Entangling class and CNOT count.* In the Weyl chamber [22] each member lies at $(a, a, 0)$ with $a = \min(|\varepsilon|, \pi/2 - |\varepsilon|)$, so the most entangling members, $|\varepsilon| = \pi/4$, are iSWAP-equivalent. Because one canonical coordinate vanishes, every non-local member needs two CNOTs rather than three [23, 24]. With the gate times above and three single-qubit layers, the target-specific synthesis time is

$$T_{\text{syn}}^{\text{tgt}}(\alpha) = \begin{cases} 0 & \alpha = \pi/2, \\ 20\,\text{ns} & \alpha \in \{0,\pi\}, \\ 150\,\text{ns} & \text{otherwise.} \end{cases} \tag{20}$$

On processors whose native entangler is itself a tunable exchange gate, as in the fSim gate set of gmon-type devices [25], each member is a single native two-qubit operation, and the synthesis overhead disappears altogether.

(iii) *Mirror symmetry.* The parity of the first mode, $P_1 = (-1)^{\hat{n}_1}$, maps $\hat{a}_1 \to -\hat{a}_1$. Conjugating Eq. (1) by P_1 therefore sends $g \to -g$ and $\phi_1 \to \phi_1 + \pi$ and leaves everything else unchanged, including the leakage structure and the boundary penalty, while on the qubits it maps $\varepsilon \to -\varepsilon$. The control problems at α and $\pi - \alpha$ are thus exactly equivalent: a pulse for one, with those two substitutions, is a pulse for the other with the same cost.

(iv) *Required exchange area.* In the qubit projection (2), the coupling alone generates Eq. (19) once $|\int g\,dt| = 2|\varepsilon|$, with g in angular units. At the 20 MHz bound this takes at most 25 ns for any member. Pulse durations are therefore set by the bandwidth filter, the boundary conditions and the cost trade-off rather than by the amount of entangling action required: each pole of the filter (12) has a time constant $1/(\pi B) \approx 32$ ns, and a filtered step needs about 52 ns to reach half its amplitude. Section VI E makes this quantitative: the largest exchange area that a bounded control can deliver through the filter in a given time is a linear program, and it sets the scale of every runtime in this paper.

B. The two sweeps

We sweep the subfamily

$$U_{\text{target}}(\alpha) = N(\alpha,\alpha,\pi/2), \qquad \alpha \in [0,\pi], \tag{21}$$

with both methods. The RL agent visits 21 targets in steps of 0.157 as a curriculum: the best checkpoint obtained at one α initializes training at the next, in the spirit of the transfer strategy of Ref. [4]. Training at each target runs in the stochastic environment at $\sigma_{\text{train}} = 1$ MHz and stops when the cost falls below 0.40 or after 150 iterations; we call a target *converged* when its final deterministic rollout reaches that cost. The direct optimizer is run independently on each of 33 targets ($\alpha = 0, 0.1, \ldots, 3.1$ and π), over horizons of 40–180 ns, with the horizon chosen by its noisy training objective. It has no memory of the other targets, so it shows how the difficulty of the control problem varies across the family, separately from what the curriculum's history adds.

C. The curriculum

Figure 4 summarizes both sweeps; the Supplemental Material [12] (Sec. S6) lists them target by target. Of the 21 curriculum targets, 14 converge. One of them is degenerate: at $\alpha = 1.570$ the target is the identity to within $|\varepsilon| = 8 \times 10^{-4}$, so the cost is already below threshold after the first 2 ns step, with the fidelity that doing nothing at all would give. It is not a fast gate, and we exclude it from every aggregate below.

The 13 non-trivial converged targets take 76–148 ns (median 132 ns), at nominal fidelity 0.982–0.992. The other 7 do not converge: the cold start $\alpha = 0$, the interval $\alpha \in [0.47, 0.79]$ and the edge $\alpha \geq 2.83$. Their final rollouts run to the 180 ns ceiling at nominal fidelity 0.92–0.95 (0.56 at $\alpha = 0$). These are failures to converge within the budget, not slow gates.

The mirror symmetry of Sec. VI A shows that the failures belong to the optimizer rather than to the targets. The curriculum grid is symmetric about $\pi/2$ to within 0.002, and in five of its ten mirror pairs one side converges while the other fails. Reflecting a converged pulse ($g \to -g$, $\phi_1 \to \phi_1 + \pi$) reproduces its cost to 10^{-12} on the mirrored target: for instance, the 118 ns pulse for $\alpha = 2.355$ implements $\alpha = \pi - 2.355$ with fidelity 0.986, where the curriculum itself ended at 180 ns and 0.949.

The curriculum runtimes are also shaped by the stopping rule. Training at each target halts as soon as the cost drops below 0.40, and every converged phase ends in the narrow band $C \in [0.393, 0.400]$: the agent has no incentive to shorten the pulse further. The runtimes are therefore times sufficient to reach a fixed cost under a bounded budget, not the shortest times the policy could achieve.

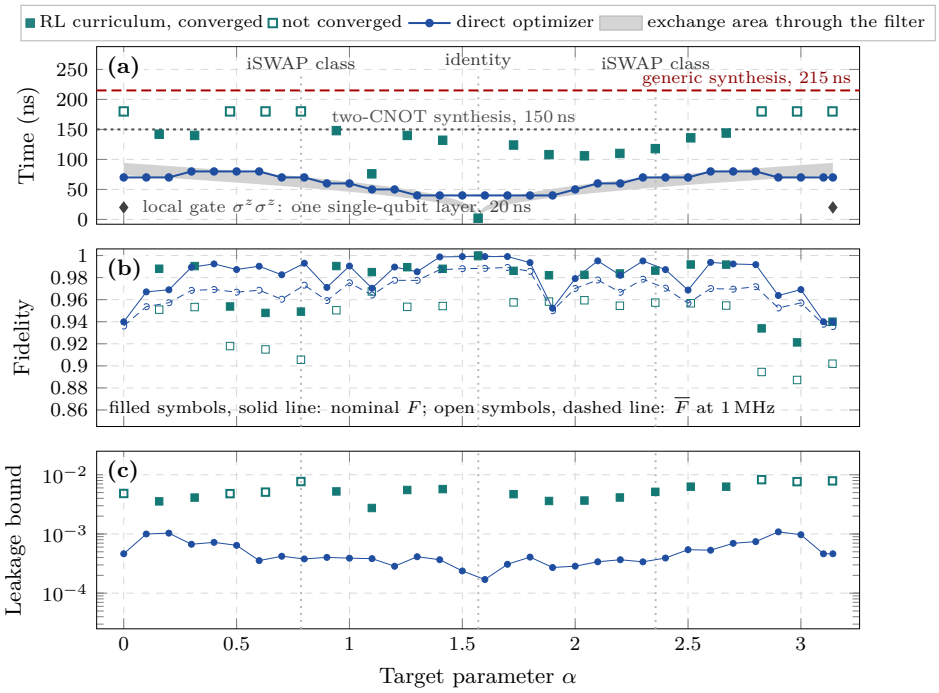


FIG. 4. Runtime across $N(\alpha, \alpha, \pi/2)$. (a) Final evaluation runtime of the RL curriculum (filled squares: converged; open squares: not converged, the rollout running to the 180 ns ceiling) and horizon selected by the direct optimizer (circles). Grey band: shortest time in which the coupling can deliver the exchange area $2|\alpha - \pi/2|$ through the bandwidth filter under the 20 MHz bound, without (lower edge) and with (upper edge) the condition that it returns to zero at the end of the pulse (Sec. VI E). Dashed: generic synthesis reference of Ref. [4], 215 ns; dotted: two-CNOT synthesis, Eq. (20), 150 ns; diamonds: the local members $\sigma_1^z\sigma_2^z$, one layer of single-qubit gates. Vertical lines mark the identity target $\alpha = \pi/2$ and the iSWAP-equivalent targets $|\alpha - \pi/2| = \pi/4$. (b) Nominal fidelity (filled symbols, solid line) and average fidelity at $\sigma_{\text{noise}} = 1$ MHz (open symbols, dashed line) for both sweeps; the cold-start curriculum point $\alpha = 0$ (0.56 and 0.61) is off scale. (c) Leakage bound of the evaluated pulses, with the symbols of panel (a).

D. Comparison with gate synthesis and with the direct optimizer

Measured against the generic reference $T_{\text{syn}} = 215$ ns, every converged non-trivial target is faster, by 1.45–2.83× (median 1.63×). This is the comparison of Ref. [4], and it needs two qualifications. First, the 180 ns episode cap lies below 215 ns, so even the failed rollouts would formally beat the reference; a speedup is meaningful only for a pulse that implements the gate. Second, the reference is generic. Against the target-specific time of Eq. (20), 150 ns for every non-local member, the converged curriculum pulses are faster by only 1.01–1.97× (median 1.14×).

The order-of-magnitude reduction reported in Ref. [4] refers to $\alpha \in [1.2, 1.7]$. By Eq. (19) this interval is the neighbourhood $|\varepsilon| \leq 0.37$ of the identity, where the required exchange area can be accumulated in about 6 ns at the coupling bound. A large ratio to a fixed three-CNOT reference is to be expected there from any working pulse optimizer; it reflects the reference more than the method. In our sweep the only target in that interval with a ratio of order ten is the identity itself.

The direct optimizer is faster than the curriculum across the family. Its horizons span 40–80 ns with a median of 70 ns, against a median of 132 ns for the converged curriculum targets, at comparable nominal fidelity (median 0.987 against 0.988) and better robustness at $\sigma_{\text{noise}} = 1$ MHz (median $\overline{F} = 0.969$ against 0.955), the latter as Eq. (18) predicts for shorter pulses. Against Eq. (20) its 31 non-local pulses are faster by 1.9–3.75× (median 2.1×), with the same range and median for the 21 of them whose nominal fidelity exceeds 0.98. Three caveats apply: the horizon grid starts at 40 ns, so the seven targets that sit at the grid minimum could be faster still; the horizon is selected by the noisy objective, which favours short pulses (Appendix B); and nine of the 33 pulses have nominal fidelity below 0.97. The runtime advantage over compiled synthesis is therefore a property of *direct analog optimization on this hardware model*, not a benefit conferred by reinforcement learning.

E. What sets the durations

Equation (19) gives the family a single physical coordinate, the exchange angle $|\varepsilon| = |\alpha - \pi/2|$ that the coupling must accumulate, and the direct optimizer follows it closely. Its horizon is 40 ns for all seven targets

with $|\varepsilon| \leq 0.33$, rises to 50–60 ns for $|\varepsilon| \in [0.37, 0.67]$ and to 70–80 ns beyond (rank correlation 0.80). It does not follow the entangling content: the iSWAP-equivalent members, the most entangling of the family, sit at 70 ns like the local ends. What varies across the family is thus not how well the target aligns with the hardware, since every member aligns perfectly, but how much exchange area the pulse must accumulate through a bandwidth-limited channel that must also start and end at zero. The favourable central region of both sweeps is the neighbourhood of the identity, not a basin of especially hardware-friendly entanglers.

This can be made quantitative. The filter (12) is linear, so the largest exchange area $\int g\,dt$ that a control bounded by 20 MHz can deliver in N steps is a linear program, with or without the condition that the filtered coupling is back at zero at the last step, which the boundary term of Eq. (7) enforces softly. Requiring the area $2|\varepsilon|$ of Sec. VI A gives the band of Fig. 4(a): from 28–40 ns at $|\varepsilon| = 0.16$ to 74–96 ns at the ends of the family. It is an estimate for pulses that use the coupling alone, not a speed limit of the model; near the ends the optimizer also uses the detunings. The horizons selected by the direct optimizer lie inside the band for 20 of the 26 targets that are not pinned at the 40 ns grid minimum. The main exceptions are the five targets with $|\varepsilon| \geq 1.43$, where the selected 70 ns is shorter than even the lower edge: there the pulses are detuning pulses whose area falls short of the π that the local gate needs, which is why their fidelity stops at 0.94–0.97. The converged curriculum pulses are 1.3–3.3 times longer than the upper edge (median 1.8). Within this model, then, the durations chosen by the direct optimizer largely track the coupling-only bound, and those of the learned pulses do not.

The same linear program applies to the single-qubit gates of the synthesis reference. A rotation by π needs a drive area $\pi/2$, which the filtered, bounded drive delivers in no less than 54 ns, or 72 ns if it must end at zero, and a $\pi/2$ rotation needs 40 and 56 ns, against the 20 ns per layer assumed in Eqs. (11) and (20). Both synthesis references therefore belong to hardware with a wider control bandwidth than the model in which the pulses are optimized, and the ratio between the two measures that mismatch as much as it measures analog control.

VII. DISCUSSION

A. The original claims and what we find

Table V sets the claims of Ref. [4] against our findings. The framework itself reproduces. Rebuilt from the published description, it produces genuine analog two-qubit gates in pulses shorter than the compiled reference; the original UFO weights sit close to the best region of an independent weight sweep, and the original network layout is as good as any we tried within the seed-to-seed spread (Sec. III E). The curriculum strategy works in the sense Ref. [4] describes: once warm-started, seven of the thirteen non-trivial converged targets were solved within 20 iterations.

The absolute fidelities are lower than those reported. Ref. [4] quotes gate infidelities of order 10^{-3} for every member of the family (its Fig. 3); our 13 non-trivial converged targets have nominal infidelities of $0.8\text{–}1.8 \times 10^{-2}$, and 7 of the 21 targets do not converge (Sec. VI). On $N(2.2, 2.2, \pi/2)$ the gap is smaller: at the lowest noise strength the average fidelity is 0.995 there, against 0.9950 and 0.9925 for our noise-free and noise-trained agents (Table II). We have not identified the cause; the training budget, the single seed and the coarser time step are candidates.

Two rows of the table need a comment.

The gradient baseline. The SGD baseline of Ref. [4] was also Adam with a noisy objective, and it received about 2000 random restarts, yet its average fidelity on this same target stays near 0.75–0.77 over the whole noise range of its Fig. 4. We cannot reconstruct why; the gap between the two gradient baselines is larger than any gap between RL and gradient optimization that we observe. Our own comparison is the least general of our findings. It sets one gradient optimizer against our RL agents, which reach cost 0.30–0.35 where the optimizer reaches 0.16 and whose curriculum fails on a third of the family; a better-trained agent could narrow the gap.

The size of the reported robustness gain. An order-of-magnitude difference in infidelity between two open-loop pulses of similar fidelity and length cannot arise under this noise model. Differences of that size can only come from different nominal fidelities or durations, or from an evaluation in which the policy acts on the noisy state (Sec. V E); Ref. [4] does not give enough detail to tell which, if either, applies to its results.

B. Comparing with the original

One point limits any quantitative comparison with Ref. [4]: the noise scales of the two studies are not the same (Sec. III B). Because the kicks are independent from step to step, the noise power per unit bandwidth is $\sigma_{\text{noise}}^2 \Delta t$, and K in Eq. (18) scales with Δt, apart from the weight of the leaking components, which grows as the step shrinks. At the step of Ref. [4], about 0.1 ns and twenty times shorter than our 2 ns, the same σ_{noise} therefore costs sixteen times less infidelity, not twenty. This alone accounts for the size of the difference between its Fig. 4, where the average fidelity of the noise-optimized controller stays between 0.995 and 0.98 over $\sigma_{\text{noise}} \in [0.1, 3.5]$ MHz, and our Table II, where that of the noise-trained agent falls from 0.9925 to 0.713. The σ_{noise} axes of the two studies should therefore not be identified (Appendix C). The conclusions above do not depend on the scale: Eq. (18) holds at every noise strength, and the matched comparison is negative over the whole range.

Other differences are what independent reimplemen-

TABLE V. The claims of Ref. [4] and what our implementation finds.

Claim	Finding	Section
The framework produces high-fidelity, low-leakage analog gates	Reproduces: nominal fidelity around 0.99 and leakage bounds of 10^{-4}–10^{-3}, with the published weights and network layout; infidelities across the family are about ten times those reported	IV, VI
RL outperforms gradient optimization, by two orders of magnitude in infidelity	Not supported: a single-start direct optimizer is ahead on every nominal metric and at every noise strength	IV, V C
Stochastic training makes the pulses robust, by up to an order of magnitude in infidelity	Not supported: no gain in four matched pairs of agents, and none is possible to leading order under white noise	V B, V D
Runtime is an order of magnitude below gate synthesis for $\alpha \in [1.2, 1.7]$	Needs a different reading: the interval surrounds the identity target and the reference is a generic three-CNOT circuit; against two-CNOT synthesis the learned pulses are 1.0–2.0× faster	VI A, VI
The runtime advantage is a benefit of RL	Not supported: the direct optimizer is faster across the family, 1.9–3.75× over two-CNOT synthesis	VI D

tation means. The two studies differ in discretization, filtering pipeline, stopping rules, curriculum schedule, training budget and evaluation protocol, and exact numerical coincidence is not the appropriate standard. The main discrepancy, however, is not an implementation detail. Two properties of the UFO formulation under white control noise follow from the formulation itself, and we would expect them in any faithful implementation: the pulse-independence of Eq. (18), and the fact that a policy observing the noisy propagator learns to respond to it rather than to produce a robust open-loop pulse. They are the reason we read the robustness results of Ref. [4] as not established, rather than as not reproduced for incidental reasons.

C. Relation to hardware

This paper compares optimizers on a common model, and none of its fidelities is a statement about a device. Three remarks place that model against current hardware (details in the Supplemental Material [12], Sec. S7). First, noise in superconducting circuits is dominated by low-frequency components, $1/f$ flux and charge noise and slow drift of the calibration [1, 26]. Of the noise models of Sec. V the quasi-static one is therefore the closer to a device, and the white model of Ref. [4], under which no training can help, the further. Second, Markovian decoherence acts like white control noise: weak relaxation and pure dephasing lower the average fidelity of any two-qubit gate of duration T by $\frac{4}{5}T(1/T_1 + 1/T_\varphi)$, whatever the gate [27], so it too leaves a pulse nothing to exploit but its duration. Third, the 10 MHz filter is narrow by hardware standards. Sycamore processors, descendants of the gmon, perform iSWAP-like gates by turning on a 20 MHz coupling for 12 ns [28], four to six times faster than the filter of the present model allows for the same exchange angle (Sec. VI E), so the runtimes of Sec. VI are set by a modelling choice of Ref. [4] more than by the coupler.

D. Limitations

The study is simulation-based. The model is truncated to three levels per mode under a rotating-wave description, and it has no transfer-function nonlinearities, spurious modes, drift or open-system decoherence. The noise used in the reproduction and in the RL training is Gaussian and white, and Eq. (18) is specific to it. We did not train an RL policy under noise with memory. Whether such a policy finds the pulse shapes that the gradient optimizer finds under quasi-static noise (Sec. V F), or better ones, is the most important open question this work leaves.

The agents of Table I, of Fig. 1, of the closed-loop test and of the runtime sweep were each trained with a single seed, in stages with changing stopping costs (Supplemental Material [12], Sec. S3); only the matched comparison was repeated with three seeds (Table III). Conclusions that rest on large effects are established: the absence of an order-of-magnitude robustness gain, the direct optimizer's lead of about 0.012 in $\overline{F}$ at 1 MHz, and the runtime results. Differences of a few 10^{-3} between two agents are not. All agents were trained at a time step of 2 ns, about twenty times coarser than that of Ref. [4], because each RL agent already takes 9.5×10^7 simulator steps; the consequences for the noise scale and for the leakage bound are discussed in Sec. VII B and Appendix B. Their training episodes almost never stopped early (Appendix B), and an implementation in which the stopping time is learned under noise could behave differently, although Eq. (18) would still bound what its pulses can gain.

The direct optimizer is deliberately minimal (one initialization and 400 iterations per horizon on a 10 ns grid starting at 40 ns) and is not a survey of gradient methods such as automatic-differentiation GRAPE or Krotov. The comparison is between one RL framework and one gradient optimizer on one hardware model, not a ranking of control methodologies, and it does not include model-based RL [11]. That gradients do well when the model is

known is not new; what this paper adds is the mechanism of Eq. (18) and the structure of the benchmark family.

E. Relation to subsequent work

Read from the present vantage point, the lasting contribution of Ref. [4] looks less like a particular optimizer and more like a demonstration that robust analog control can be posed as a learning problem with leakage-aware costs and stochastic perturbations. The literature since has refined that in two directions. Model-free and hardware-interacting schemes reduce model bias by adapting on the device itself [10], and model-based or hybrid methods address the sample inefficiency that makes pure model-free RL expensive [11]. Our results sharpen the case for both. When the model is known and the noise is white, a direct optimizer wins both the nominal and the robustness problem, simply by producing short and accurate pulses. What a noise-trained policy does learn is feedback, which is useful only where the state can be observed during the operation. Learning therefore earns its cost where the model is uncertain or the noise has structure that a pulse can exploit: the regime of closed-loop refinement against the device, where the perturbation distribution is the device's own rather than a modelled one [29–31].

VIII. CONCLUSION

We have independently reimplemented the UFO framework for robust analog two-qubit gate design on a multilevel gmon model and compared it with a direct gradient optimizer that uses the same Hamiltonian, the same objective and the same training noise. The framework works: it produces high-fidelity analog gates, faster than a compiled implementation, and its published hyperparameters are close to optimal in our hands. Its headline claims are not supported in our implementation. The RL agent does not outperform a direct optimizer, training under noise does not make the deployed pulses more robust, and the order-of-magnitude runtime gain sits around a target that is the identity.

Two results explain these findings and should transfer beyond this study. Against temporally uncorrelated control noise, the open-loop robustness of any pulse is fixed to leading order by its nominal fidelity and its duration, Eq. (18), with a rate that we compute from the Hamiltonian and verify on every pulse we produced; no training procedure can improve on that except by making pulses more accurate or shorter. And every member of the benchmark subfamily $N(\alpha, \alpha, \pi/2)$ is an exchange rotation by $\alpha - \pi/2$, which makes its runtime landscape a function of the exchange angle and fixes the fair synthesis reference at two CNOTs.

The useful question for noise-aware learning in quantum control is therefore not how large its advantage is under white noise (to leading order, there is none) but whether it finds pulse shapes that exploit noise with memory, as a gradient optimizer trained against quasi-static noise already does to a modest degree, and whether the robustness it reports is that of a deployable open-loop pulse or of a feedback policy. Benchmarks that separate those two cases, and that state the time step and units of their noise model, would make results in this area comparable.

Appendix A: First-order response to control noise

Let the noise on channel c take the value $\xi_{c,k}$ (in MHz) during step k, so that it adds $s\,\xi_{c,k}\hat{G}_c$ to the Hamiltonian $\hat{H}_k$ of that step, with $s = 2\pi \times 10^{-3}$ and $\hat{G}_c$ the operator that the channel multiplies in Eq. (1). To first order the noisy propagator is $U = U_0\big[1 - is\sum_{c,k}\xi_{c,k}\tilde{G}_{c,k}\big]$, with the toggling-frame generators

$$\tilde{G}_{c,k} = W_k^\dagger\Big[\int_0^{\Delta t} e^{i\hat{H}_k\tau}\hat{G}_c\,e^{-i\hat{H}_k\tau}d\tau\Big]W_k, \tag{A1}$$

where U_0 is the nominal propagator and W_k its part up to the start of step k. For zero-mean Gaussian noise with covariance $\langle\xi_{c,k}\xi_{c',l}\rangle = \delta_{cc'}C(k,l)$, and a nominal gate that reaches its target, the average fidelity (14) is, to leading order,

$$\begin{aligned} 1-\overline{F} &\simeq 1-\overline{F}(0) + \frac{d}{d+1}\,s^2\sum_c\sum_{k,l}C(k,l)\,v_c(k,l),\\ v_c(k,l) &= \frac{\mathrm{Re}\,\mathrm{Tr}\big(P\tilde{G}_{c,k}\tilde{G}_{c,l}P\big)}{d}\\ &\quad - \frac{\mathrm{Tr}\big(P\tilde{G}_{c,k}P\big)\mathrm{Tr}\big(P\tilde{G}_{c,l}P\big)}{d^2}. \end{aligned} \tag{A2}$$

For stationary noise the double sum is the overlap of the noise spectrum with the filter function $\Phi_c(f) = \sum_{k,l}v_c(k,l)\cos[2\pi f(t_k - t_l)]$ of the pulse [17–19]. White noise, $C(k,l) = \sigma^2_{\mathrm{noise}}\delta_{kl}$, keeps only the diagonal $v_c(k,k)$, which is the variance V of Sec. V B evaluated on $\tilde{G}_{c,k}$ and does not depend on W_k as long as the nominal evolution keeps the computational subspace invariant: this is Eq. (18), and in the frequency domain it is the statement that the average of Φ_c over the Nyquist band is the same for every pulse of a given duration. Quasi-static noise, $C(k,l) = \sigma^2_{\mathrm{noise}}$, keeps the whole sum, $\Phi_c(0)$, in which generators at different times can add or cancel; that is where the pulse shape enters.

We evaluate Eq. (A1) exactly, in the eigenbasis of each $\hat{H}_k$, and check Eq. (A2) against Monte Carlo replays of the full dynamics with noise of the same covariance. Table VI gives the white-noise case channel by channel.

TABLE VI. White-noise infidelity rate per channel, in $10^{-5}\,\mathrm{ns}^{-1}\,\mathrm{MHz}^{-2}$: first-order value of Eq. (A2) and Monte Carlo replay with the noise on that channel alone ($\sigma_{\mathrm{noise}} = 1\,\mathrm{MHz}$, 4000 realizations). Ranges are over the four pulses of Fig. 1 and the 104 ns pulse of Sec. IV.

Channel	First order	Monte Carlo
Detuning δ_1 or δ_2	1.579–1.580	1.50–1.59
Drive amplitude f_1 or f_2	9.927–9.934	9.66–10.04
Coupling g	6.772–6.775	6.47–7.03
Anharmonicity η	< 0.002	< 0.002
All channels	29.79–29.80	28.8–29.3

Appendix B: The noisy leakage bound

In our implementation every term of Eq. (7) is evaluated on the noisy trajectory, including the leakage bound, whose second time derivative responds to the step-to-step jumps of white noise. This changes the objective that a noise-trained method sees. At $\sigma_{\mathrm{train}} = 1\,\mathrm{MHz}$ the bound grows by 3.8×10^{-3} per ns for every pulse we replayed, about 27 times the leakage population actually present at the end of the pulse; for the 116 ns pulse of the noise-free agent the term βL_{tot} is 4.5, against 0.50 for $\chi(1-F)$. In the noisy objective, leakage thus acts as an implicit runtime penalty about 23 times stronger than the explicit one, κ/T_{norm}, and one that is nearly blind to pulse shape. This explains two observations. The horizon search of the direct optimizer, whose sampled losses rise by about 0.04 per ns from 60 ns on, prefers the 60 ns pulse (fidelity 0.982, cost 0.381) to the more accurate 70 ns one. And the cost of a noisy RL training rollout never falls below the stopping threshold: all 1.05×10^6 training episodes of the noise-trained agent ran to the 180 ns ceiling, so the co-optimization of stopping time that distinguishes the RL formulation was inactive throughout noisy training.

The bound is not the only reason. In the noise-trained agents of pairs 2–4 (Table III), whose bound does not see the noise, the training episodes still lasted 180.0 ns on average, and those of the noise-free agents 177–178 ns: the exploration noise of the stochastic policy, with the training noise on top of it, keeps the cost of a training rollout above a threshold that the deterministic pulse of the same policy reaches at about 100 ns.

The effect depends strongly on the time step. The part of the bound that responds to white noise contains a fourth finite difference of the controls and scales as Δt^{-4}: replayed on finer grids, its growth of 3.8×10^{-3} per ns at 2 ns becomes about 15, 250 and 4200 times larger at 1, 0.5 and 0.25 ns. On a grid of a thousand steps a bound accumulated from the noisy Hamiltonian would exceed every other term of the objective by orders of magnitude, so Ref. [4] presumably evaluates it otherwise; it does not say how. The domination of the noisy objective by the bound is therefore a property of our implementation, and we cannot tell whether it applies to theirs. The direct-optimizer test of Sec. V F and the matched comparison of RL agents (Table III) were both repeated without it, and their outcome did not change.

The direct optimizer with the bound taken from the noise-free controls. We repeated the experiment of Table IV with the bound taken from the noise-free controls, so that the noise enters the objective through the fidelity alone. The cost of white-noise training in nominal fidelity falls from 4.8, 5.1 and 10.1×10^{-3} to 2.0, 3.5 and 8.1×10^{-3} at the three horizons, and the outcome is the same: again none of the 24 pulses is more accurate, or more robust against white noise, than the one optimized without noise. Over the two variants white-noise training helped in none of 48 runs. The quasi-static results change by less than 10^{-5} in every fidelity.

Appendix C: Noise scales and time step

This appendix gives the details behind Sec. VII B. The same σ_{noise} describes a different physical noise level at a different time step, and it matters whether amplitudes are quoted in cyclic or angular units. Replaying our pulses with the noise redrawn every 0.1, 0.5, 1, 4 or 8 ns instead of every 2 ns, at the same σ_{noise}, multiplies the excess infidelity by 0.06, 0.31, 0.6, 1.3 and 2.5, as the first-order rate predicts to within 5%; the factor is not exactly the ratio of the steps because the components of the noise that lead out of the subspace weigh more at a finer step. It also matters where the noise enters. Added to the raw controls, before the bandwidth filter instead of after it, the same σ_{noise} costs our five pulses 5 to 18 times less infidelity, and reaches the Hamiltonian with the 32 ns memory of the filter. In physical terms, the detuning noise of our model at $\sigma_{\mathrm{noise}} = 1\,\mathrm{MHz}$ and $\Delta t = 2\,\mathrm{ns}$ is white frequency noise that dephases each qubit in $T_\varphi = 2/[(2\pi\sigma_{\mathrm{noise}})^2 \Delta t] \approx 25\,\mu\mathrm{s}$, and 3.5 MHz corresponds to $2\,\mu$s.

Ref. [4] describes its pulses as piecewise constant with "around one thousand time steps" each, which for gates of about 100 ns is a step of about 0.1 ns, twenty times shorter than ours; it does not state its convention for σ_{noise}. At that step the same σ_{noise} costs sixteen times less infidelity, and the fidelity loss that its Fig. 4 shows over $\sigma_{\mathrm{noise}} \in [0.1, 3.5]\,\mathrm{MHz}$ is indeed more than an order of magnitude smaller than ours for pulses of comparable length: there the average fidelity of the noise-optimized controller stays between 0.995 and 0.98, while ours falls from 0.9925 to 0.713 (Table II). Dividing our loss by sixteen gives about 0.991 at 1 MHz, so the time step alone accounts for a difference of that size; angular units or noise injected before the filter would change it further. The σ_{noise} axes of the two studies should not be identified, and no number in this paper at a given σ_{noise} should be compared with a number at the same σ_{noise} in Ref. [4].

The same applies to the training noise. Under the same assumptions, $\sigma_{\mathrm{train}} = 1\,\mathrm{MHz}$ at 2 ns corresponds to about 4 MHz at 0.1 ns, above the range evaluated in Ref. [4]. We did not train at the matched weaker level;

by Eq. (18) the pulse-independence does not depend on it, but the quality of the trained agent may.

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